

Democratizing Sprint Training with a Low-Cost Reaction-Time and Photo-Finish System for Track & Field

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Abstract

Non-elite athletes and coaches can rarely measure block-start reaction time with millisecond accuracy. Sprint medals are decided by milliseconds, yet the systems that time reaction and finish are costly [1], and only federations and large clubs own them. To bridge this gap, we present a low-cost, portable start-and-timing system that retrofits onto existing starting blocks, generating the start sequence and computing reaction time from an on-block accelerometer, connected to a phone-camera photo finish. Across 79 block starts, raising the detection threshold from 10 to 50 mg delays a single-threshold detector by a median of 9.3 ms, whereas our two-stage detector's reaction time remains unaffected (median shift 0.0 ms). Its random error is estimated at 1–2 ms; reaction timing will be validated against a mechanical ground truth, and the photo finish against a 2000 fps camera.

1 Introduction

In competitive sprinting, elite athletes train for years just to gain a few hundredths of a second: Noah Lyles won the Paris 2024 Olympic 100 m final by five thousandths [2]. Reaction times differ by several hundredths of a second between elite sprinters [3], so a large part of the winning margin can be decided before the athlete has even moved. Reaction time is the interval from the starting signal to the first force on the block; a start under 0.100 s is a false start, a limit based on the assumed minimum auditory reaction time [4], and the fastest valid reactions of Olympic sprinters lie only slightly above it [3, 5]. The automatic systems that resolve it are installed at certified competitions [6], and even a single-block training version lists at about US\$1,600 [1]. Thus, coaches rely on hand timing, which reads about 0.24 s faster than electronic timing [7], an error far larger than the gains an athlete trains for.

Portable tools such as photocells begin timing when the athlete's torso crosses a beam, so their times exclude the reaction to the gun [8]. Certified systems such as FinishLynx ReacTime measure reaction time to 1/1000 s from sensors on every block and integrate with photo finish cameras [6]: the ReacTime patent describes detecting the onset when an acceleration signal rises over a threshold [9], which by construction reports the onset only after it has begun. By contrast, the Omega system used at the Olympic Games measures the force against the block's footrest 4,000 times per second and pairs it with a finish camera of up to 40,000 images per second [10]. Earlier student work took the same force-based approach with load cells, verified for operational robustness (18 of 20 starts) but without a formal timing-accuracy characterization [11]. However, ReacTime is costly [1], and force-based systems such as Omega's rely on sensors built into the

blocks, leaving a clear gap for a low-cost, portable solution with a documented timing error budget.

The contributions of this paper are a low-cost start unit, a small box that mounts on any block, and the measurements that characterize it. Its two-stage onset detector, an STA/LTA trigger [12] followed by Maeda's AIC picker [13], keeps the onset stable as the trigger threshold changes, and the timing path gives a random error of 1–2 ms, dominated by the accelerometer's sampling interval. A phone photo finish, linked by radio, is planned to time the finish to within one video frame, at worst 33 ms at 30 fps. Validating the system is not trivial: a commercial system is not necessarily more precise than our picker, and a threshold detector reads the onset later by construction [9]; we therefore aim to validate against a mechanical ground truth.

2 Design

Architecture. Our system consists of a start unit on the block, which gives the start signal and times the reaction, and a phone at the finish line, which records the finish (Figure 1).

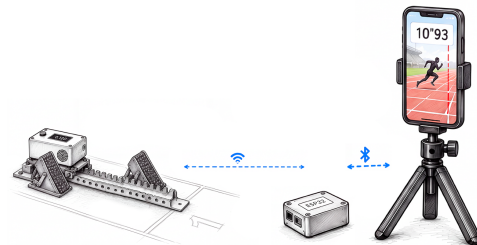


Figure 1. System overview (AI-generated).

The start unit is built on the Seeed XIAO nRF52840 Sense, a board with an on-module LSM6DS3TR-C accelerometer [14], and sits on the back of the block with a button and a buzzer (first prototype in Fig-

ure 2; Prototype design in Appendix B). It sends the start instant through a wired interrupt to an ESP32, which relays it over long-range ESP-NOW to a second ESP32 at the finish line, connected to the phone via BLE. If the link drops, a store-and-forward sync between the start unit and the phone acts as a fallback. Only the start unit is built and evaluated here.

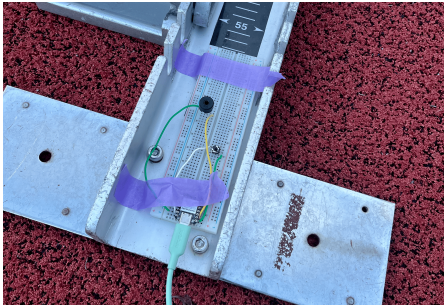


Figure 2. First prototype on breadboard.

Start sequence and detection. A button press starts three beeps (“on your marks”, “set”, “go”), each stamped by an 8 μ s micros() counter, so no latency enters. After “set” the unit waits until the sprinter is still (horizontal acceleration under 15 mg for 200 ms) and fires “go” 0.7–1.5 s later at random. The accelerometer runs into a RAM ring at 865 Hz measured (833 Hz nominal), which sets the picker’s resolution to one sample, 1.2 ms; gravity is estimated at rest as a vector, so each sample splits into vertical and horizontal parts regardless of the mounting angle. Onset detection runs in two stages: an STA/LTA trigger first decides that a movement occurred, without fixing exactly when; an AIC picker then finds, inside a 200 ms window around that trigger, the sample that best splits a quiet segment from an active one, and reports it as the onset. This step is threshold-free, so the reported time stays stable however the trigger threshold is set. An onset under 100 ms after “go” is a false start. the source code can be consulted in Appendix A.

Photo finish. The mobile app opens the camera and records continuously; once it receives the start instant from the start unit over the connection (after clock synchronization), on-device AI detects the athlete’s torso crossing the finish line in the video, and the finish time follows directly as $t_{\text{end}} - t_0$.

3 Evaluation

Method. Five sprinters perform 79 attempts on different blocks over four sessions. The prototype is mounted on the back of the block; attempts are dumped over USB to CSV with the firmware’s own set, arming and go stamps. The board samples at 865 Hz without gaps, with 334 μ s timestamp jitter (1σ), so the random error is 1–2 ms, set by the picker’s one-sample resolution (1.2 ms).



Figure 3. Block start after “go” (green): raw acceleration, horizontal magnitude and STA/LTA ratio, with the AIC onset (black) at 138 ms vs. threshold-only (dashed).

Onset picker. On the recorded start of Figure 3, disabling the AIC step shifts the reported time by 16.2 ms. When we repeat the threshold sweep (10–50 mg) on all 79 starts, the AIC onset stays identical in 70 starts and within one sample (1.2 ms) in 74; without AIC, it moves by a median of 9.3 ms (Figure 4). In the remaining 5 starts the threshold changes which movement is picked as the first one, not just its timing. This shows the result is reproducible, not correct: even ReacTime’s own patent calls its threshold “illustratively at about 1.5 g” [9], a design choice, not a standard, so matching it would prove nothing but shared convention.

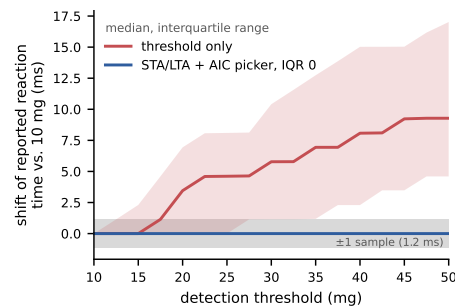


Figure 4. Threshold sweep, 79 starts: the AIC onset stays fixed; threshold-only drifts up to 9.3 ms (median).

Real starts. Across 79 attempts over four sessions, the unit records 72 valid starts (median reaction 154.0 ms, range 111.3–476.5 ms) and 7 false starts. While the sprinter holds “set”, the block reads only a few mg, against push-off peaks of 1–3.6 g.

4 Conclusions

We built a low-cost box that clips onto any starting block and times a sprinter’s reaction independently of its detection threshold, closing the gap between elite-grade timing and what non-elite clubs can afford. The result that matters is practical: an athlete training alone gets a stable, millisecond-resolution reaction time, at a fraction of the cost. Next, we validate it against a mechanical ground truth, then add the ESP-NOW link to the phone photo finish.

References

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A Source code

The firmware and analysis code are available at <https://github.com/lorenzogalli-dev/Reaction-Time-System>.

B Prototype design

Figure 5 shows the Prototype design of the start unit.

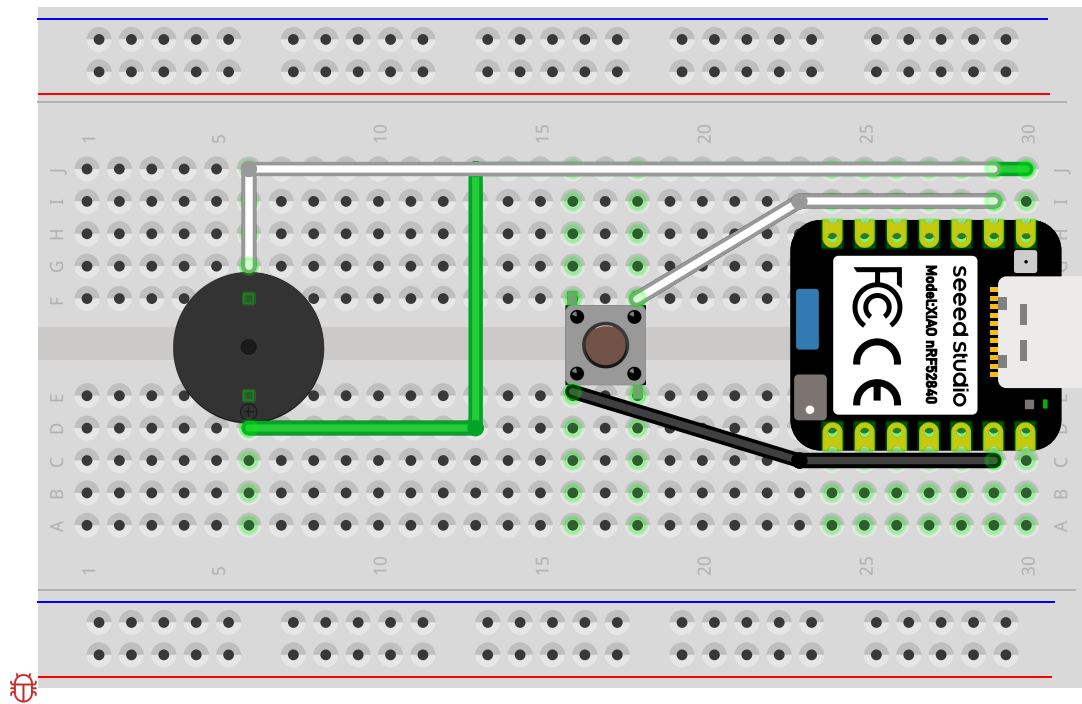


Figure 5. Prototype design of the start unit.